Harkamal Singh

DATA SCIENTIST

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Junior Data Scientist with 1.5+ years of hands-on project work in Python, SQL, statistics, and machine learning. Experienced in designing experiments, building reproducible ML systems, and translating model results into actionable decisions. Contributed three merged pull requests to established open-source Python projects.

PROFESSIONAL EXPERIENCE

Open-Source Software Contributor | Independent | Remote | Oct 2026 - Present

- Delivered 3 merged, maintainer-reviewed pull requests across aeon-neuro and Kedro Plugins, improving testing, public package APIs, and dataset documentation.
- Corrected 8 estimator parameter hooks and exported ChannelFilter; contributed 12 new test cases and validated 27 channel-selection tests locally.
- Used pytest, Git, pre-commit, and GitHub Actions to validate changes and address real issues in established Python repositories.

TECHNICAL SKILLS

Programming & Analytics: Python, SQL, pandas, NumPy, SciPy, EDA, feature engineering, data visualization

Statistics & Machine Learning: A/B testing, causal inference, uplift modeling, time series, scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch

Databases & Tools: MySQL, MongoDB, Redis, DuckDB, Streamlit, Tableau

Engineering & MLOps: Git, GitHub Actions, pytest, FastAPI, Docker, MLflow, ZenML, CI/CD

SELECTED PROJECTS

Movie Recommendation A/B Testing & Uplift Modeling | Python, SciPy, scikit-learn | GitHub

- Designed randomized experiment for 28,000 synthetic users; measured +0.846 percentage-point CTR lift ($p = 0.017$) and recommended personalization for returning users only.

LocalMart AI - Retail Promotion Intelligence | CatBoost, causal ML, PuLP | GitHub

- Combined demand forecasting, treatment-effect estimates, product economics, and constrained optimization; estimated $20.6K simulated incremental profit in a semi-synthetic setting.

Flight Pulse - AI Flight Delay Analyst | Python, SQL, MySQL, Gemini | GitHub

- Integrated 72 Toronto Pearson flight records with hourly weather context; built 7 approved read-only AI analyst tools and 70 automated tests.

End-to-End House Price MLOps System | ZenML, MLflow, FastAPI, Docker | GitHub

- Built ML pipelines, model monitoring, promotion and rollback controls, and CI; achieved test $R^2 = 0.9221$ with 28 automated tests.

Fighting Churn - Retention Risk Modeling | Python, XGBoost, SQL | GitHub

- Developed churn risk ranking and time-based evaluation; achieved 2.07x lift among the highest-risk 10% compared with random targeting.

Toronto Climate & Time-Series Analysis | Python, statistics, time series | GitHub

- Analyzed Toronto temperature records from 1953-2023 and estimated annual mean warming of +0.218°C per decade.

EDUCATION

Conestoga College | Diploma, Computer Programming | Waterloo, ON | Sep 2024 - Apr 2026

Relevant Coursework: Data Structures, Object-Oriented Programming, Databases, Software Development, AI & ML, Statistics & Probability